

Data Cleaning & Reporting Automation

Automated pipeline that cleans messy sales data and generates a formatted Excel report with live formulas and charts.

Files

File	Description
<code>raw_sales_data.csv</code>	Sample messy input data (503 rows) – mixed date formats, inconsistent casing, missing values, duplicates,

File	Description
	price outliers
<code>clean_pipeline.py</code>	Reusable Python cleaning script
<code>cleaned_sales_data.csv</code>	Output of the pipeline (480 rows)
<code>cleaned_sales_data_log.json</code>	Machine-readable log of every change made
<code>Sales_Data_Report.xlsx</code>	Auto-generated report: summary KPIs, cleaned data table, pivots,

File	Description
	charts, cleaning audit log

How to run it on your own data

```
pip install pandas numpy openpyxl  
python clean_pipeline.py  
your_file.csv cleaned_output.csv
```

Works with `.csv`, `.xlsx`, or `.xls` input.

Produces `cleaned_output.csv` and
`cleaned_output_log.json`.

To regenerate the Excel report from a cleaned file, run `build_report.py` (reads `cleaned_sales_data.csv` + the log JSON, writes `Sales_Data_Report.xlsx`).

What the pipeline does

1. **Standardize formats** — trims whitespace, normalizes text casing, parses inconsistent date formats into one format

2. **Handle missing values** — numeric columns filled with median, categorical columns filled with "Unknown" (each fill is logged)
3. **Remove duplicates** — drops exact duplicate rows and duplicate keys (e.g. repeated Order IDs)
4. **Recompute derived fields** — e.g. Revenue = Quantity × Unit Price, recalculated after cleaning
5. **Flag outliers** — IQR-based flagging on price, added as a boolean column rather than silently dropped
6. **Log everything** — every transformation is written to a JSON log for auditability

Customizing

- **Missing value strategy:** edit `handle_missing()` in `clean_pipeline.py` — swap median for mean, a fixed value, or a drop-row rule
- **Outlier column:** change the `col` argument in `flag_outliers()` to target a different field

- **Duplicate key:** change `key_cols` in `remove_duplicates()` to match your dataset's unique identifier
- **Report layout:** `build_report.py` uses `openpyxl`; add/remove sheets or charts as needed

Report contents

(`Sales_Data_Report.xlsx`)

- **Summary** — KPIs (rows cleaned, duplicates removed, missing values fixed, outliers flagged, total revenue) as live formulas
- **Cleaned Data** — full dataset as a formatted Excel table
- **Breakdowns** — region/category pivots via `SUMIF` / `COUNTIF` , plus bar, pie, and line charts
- **Cleaning Log** — full audit trail of every transformation applied

All formulas recalculate automatically when source data changes; 497 formulas verified with zero errors.